Homework 01: Solutions

REAL VARIABLES I

Fall 2026

Solutions for Homework 01.

1. (Monotone convergence for nonnegative sums)

1.a

Remark (Minimal solution). Let $S_k \coloneqq \sum_{j\in\mathcal{J}} a_{j,k}$ and $S \coloneqq \sum_{j\in\mathcal{J}} a_j$. By monotonicity, $S_k \le S_{k+1} \le S$. Therefore $\lim_k S_k$ exists and $\lim_k S_k \le S$.
Next, we show that $\lim_k S_k \ge S$. Fix an arbitrary $c < S$; it suffices to show that $S_k \ge c$ for all k large enough. Choose a finite $F \subseteq \mathcal{J}$ with $\sum_{j\in F} a_j > c$. By algebra of limits, $\sum_{j\in F} a_{j,k} \to \sum_{j\in F} a_j > c$. Therefore $S_k \ge \sum_{j\in F} a_{j,k} > c$ for all k large enough. Hence $\lim_k S_k \ge c$, as desired.

Remark (Solution). Let $S_k \coloneqq \sum_{j\in\mathcal{J}} a_{j,k}$ and $S \coloneqq \sum_{j\in\mathcal{J}} a_j$. We show that $S_k \uparrow S$.
Step 1: (S_k) is nondecreasing and therefore $\lim_k S_k$ exists. Since $a_{j,k} \le a_{j,k+1}$ for all j, monotonicity gives $S_k \le S_{k+1} \le S$. A nondecreasing sequence in $[0,+\infty]$ converges to its supremum, so $\lim_k S_k$ exists.
Step 2: $\lim_k S_k \le S$. Since $a_{j,k} \le a_j$ for all j, monotonicity gives $S_k \le S$ for every k. Therefore $\lim_k S_k \le S$.
Step 3: $\lim_k S_k \ge S$. If $S = 0$ this holds because $\lim_k S_k \ge 0$. Otherwise it suffices to show that for every real c with $0 \le c < S$ it holds that $S_k \ge c$ for all large enough k. Fix such a c.
By the definition of S as a supremum of finite sums there is a finite $F \subseteq \mathcal{J}$ with $\sum_{j\in F} a_j > c$; this F depends only on c. For each of the finitely many $j \in F$, the sequence $a_{j,k}$ is nondecreasing with limit $a_j \in [0,+\infty]$. Hence, by the algebra of limits (valid in $(-\infty,+\infty]$ for finite sums of nondecreasing sequences), $\sum_{j\in F} a_{j,k} \to \sum_{j\in F} a_j > c$.
Hence there is K, depending only on F and c, with $\sum_{j\in F} a_{j,k} > c$ for all $k \ge K$. For such k, the definition of S_k gives $S_k \ge \sum_{j\in F} a_{j,k} > c$, as desired.

1.b

Remark (Solution). Let $a_{j,k} \coloneqq 1$ if $j = k$ and $a_{j,k} \coloneqq 0$ otherwise. We show that both limits exist and $\lim_k \sum_j a_{j,k} = 1 \ne 0 = \sum_j \lim_k a_{j,k}$.
For fixed j, $a_{j,k} = 0$ for all $k > j$. Hence $\lim_k a_{j,k} = 0$, and so $\sum_j \lim_k a_{j,k} = 0$. For fixed k, the only nonzero term of $(a_{j,k})_j$ is $a_{k,k} = 1$. Hence $\sum_j a_{j,k} = 1$ for every k, and so $\lim_k \sum_j a_{j,k} = 1$.

Remark (Discussion). The mass 1 sits at position $j = k$ and escapes to infinity as k grows, so it is invisible in each coordinate limit. Monotone convergence fails because $a_{j,k}$ is not nondecreasing in k; see also Problem 2(b) for why no summable dominating profile can exist.

2. (Dominated convergence for nonnegative sums)

2.a

Remark (Minimal solution). Since $\sum_j g_j < \infty$, each g_j is finite. Since $a_{j,k} \le g_j$, we get $a_j = \lim_k a_{j,k} \le g_j$. Fix N and split every series at N (a nonnegative sum equals its series),

$$\Big|\sum_{j\in\mathbb{N}} a_{j,k} - \sum_{j\in\mathbb{N}} a_j\Big| \le \sum_{j\le N} |a_{j,k} - a_j| + |\sum_{j>N} a_{j,k} - \sum_{j>N} a_j|.$$

Using the triangle inequality and monotonicity

$$|\sum_{j>N} a_{j,k} - \sum_{j>N} a_j| \le |\sum_{j>N} a_{j,k}| + |\sum_{j>N} a_j| \le 2\sum_{j>N} g_j.$$

Therefore

$$\Big|\sum_{j\in\mathbb{N}} a_{j,k} - \sum_{j\in\mathbb{N}} a_j\Big| \le \sum_{j\le N} |a_{j,k} - a_j| + 2\sum_{j>N} g_j.$$

By algebra of limits, the finite sum tends to 0 as $k \to \infty$. Therefore

$$\limsup_{k\to\infty} \Big|\sum_{j\in\mathbb{N}} a_{j,k} - \sum_{j\in\mathbb{N}} a_j\Big| \le 2\sum_{j>N} g_j.$$

Since the LHS does not depend on N, take the limit $N \to \infty$ to obtain

$$\limsup_{k\to\infty} \Big|\sum_{j\in\mathbb{N}} a_{j,k} - \sum_{j\in\mathbb{N}} a_j\Big| = 0,$$

as desired.

Remark (Solution). We show that $\lim_k \sum_{j\in\mathbb{N}} a_{j,k} = \sum_{j\in\mathbb{N}} a_j$.

Step 1: $a_{j,k} \le g_j < \infty$ and $a_j \le g_j$. Since $\sum_j g_j < \infty$, every g_j is finite. Hence $a_{j,k} \le g_j < \infty$. Passing to the limit in k gives $a_j \le g_j$.

Step 2: a bound independent of k. A nonnegative sum over $\mathbb{N}$ equals the corresponding series. Therefore all sums below are series of finite nonnegative terms, and for every $N \in \mathbb{N}$

$$\sum_{j\in\mathbb{N}} a_{j,k} - \sum_{j\in\mathbb{N}} a_j = \sum_{j\le N} (a_{j,k} - a_j) + \Big(\sum_{j>N} a_{j,k} - \sum_{j>N} a_j\Big).$$

By Step 1 and monotonicity, the tails $\sum_{j>N} a_{j,k}$ and $\sum_{j>N} a_j$ are nonnegative and at most $\sum_{j>N} g_j$. Hence, by the triangle inequality,

$$|\sum_{j>N} a_{j,k} - \sum_{j>N} a_j| \le \sum_{j>N} a_{j,k} + \sum_{j>N} a_j \le 2\sum_{j>N} g_j.$$

Therefore, again by the triangle inequality,

$$\Big|\sum_{j\in\mathbb{N}} a_{j,k} - \sum_{j\in\mathbb{N}} a_j\Big| \le \sum_{j\le N} |a_{j,k} - a_j| + 2\sum_{j>N} g_j.$$

Step 3: conclusion. Fix N. The finite sum in Step 2 has finitely many terms, each tending to 0 as $k \to \infty$. Hence it tends to 0, and therefore $\limsup_k |\sum_j a_{j,k} - \sum_j a_j| \le 2\sum_{j>N} g_j$. The left side does not depend on N. Moreover, $\sum_{j>N} g_j \to 0$ as $N \to \infty$, because it is the tail of a convergent series. Therefore the lim sup is 0, which is the claim.

2.b

Remark (Solution). Take $a_{j,k} \coloneqq 1$ if $j = k$ and 0 otherwise, as in Problem 1(b). By Problem 1(b), the limits $a_j = \lim_k a_{j,k} = 0$ and $\lim_k \sum_j a_{j,k} = 1$ exist. Moreover $1 \neq 0 = \sum_j a_j$.

Remark (Discussion). Domination fails here: any profile with $a_{j,k} \leq g_j$ for all k needs $g_j \geq a_{j,j} = 1$ for every j, so $\sum_j g_j = \infty$.

3. (Countable sets have null Lebesgue outer measure)

3.a

Remark (Solution). Let $E \subseteq \mathbb{R}$ be countable. We show that $\mathcal{L}^*(E) \leq \epsilon$ for every $\epsilon > 0$. Since $\mathcal{L}^*(E) \geq 0$ (every sum in the infimum has nonnegative terms), this gives $\mathcal{L}^*(E) = 0$.
Choose a sequence $(e_j)_{j\geq 0}$ with $E \subseteq \{e_j : j \geq 0\}$ (list E with repetitions if it is finite, and take $e_j = 0$ for all j if $E = \emptyset$). Fix $\epsilon > 0$. Then $E \subseteq \bigcup_{j\in\mathbb{N}} (e_j - \epsilon 2^{-j-2}, e_j + \epsilon 2^{-j-2})$, because every element of E equals some e_j. Therefore, $\left((e_j - \epsilon 2^{-j-2}, e_j + \epsilon 2^{-j-2})\right)_{j\in\mathbb{N}}$ is a countable cover of E by open intervals with total length

$$\sum_{j\in\mathbb{N}} 2\epsilon 2^{-j-2} = \sum_{j\in\mathbb{N}} \epsilon 2^{-j-1} = \epsilon,$$

and so $\mathcal{L}^*(E) \leq \epsilon$.

4. (Countable and uncountable sets)

4.a

Remark (Solution). For $N \in \mathbb{N}$ let $S_N \coloneqq \{(a_n) \in \mathbb{Q}^{\mathbb{N}} : a_n = 0 \text{ for all } n > N\}$. We show that $\ell_c(\mathbb{N}; \mathbb{Q})$ is countable.
Every sequence with finite support lies in S_N for N the largest index of its support (take $N = 0$ if the support is empty). Hence $\ell_c(\mathbb{N}; \mathbb{Q}) = \bigcup_{N=0}^{\infty} S_N$. The map $(a_n) \mapsto (a_0, \dots, a_N)$ is a bijection $S_N \to \mathbb{Q}^{N+1}$. Its inverse is $(q_0, \dots, q_N) \mapsto (q_0, \dots, q_N, 0, 0, \dots)$, because the entries of an element of S_N with $n > N$ vanish. $\mathbb{Q}^{N+1}$ is countable as a finite product of countable sets, and hence so is S_N. Therefore $\ell_c(\mathbb{N}; \mathbb{Q})$ is a countable union of countable sets, so it is countable.

4.b

Remark (Solution). We show that there are injections $\mathbb{R} \to \mathbb{R}^2$ and $\mathbb{R}^2 \to \mathbb{R}$; the Cantor–Bernstein theorem then gives $|\mathbb{R} \times \mathbb{R}| = |\mathbb{R}|$.
The map $x \mapsto (x, 0)$ is an injection $\mathbb{R} \to \mathbb{R}^2$.
Step 1: Reduction to $(0, 1)$**.** $\phi(x) \coloneqq \frac{1}{2} + \frac{1}{\pi} \arctan x$ is strictly increasing, hence a bijection of $\mathbb{R}$ with its range, $(0, 1)$.
It is therefore sufficient to prove that $(0, 1)^2$ can be injected into $(0, 1)$.
Step 2: Reduction to decimal expansions. Let $D = \{s \in \{0, 1, \dots, 9\}^{\mathbb{N}} : s_n \neq 9 \text{ for arbitrarily large } n\}$. This is the set of decimal expansions $s \leftrightarrow \sum_{n\in\mathbb{N}} s_n 10^{-n-1}$ of numbers in $[0, 1)$ (indexed by $n \in \mathbb{N} = \{0, 1, \dots\}$, like all sequences in this solution). Let $D_0 = D \setminus \{(0, 0, \dots)\}$. We use that $(0, 1)$ is in bijection with D_0.
Therefore it is sufficient to show that D_0^2 can be injected into D_0.
Step 3: Conclusion via interleaving.
For every $s, t \in D_0$ define $\iota(s, t) = (s_0, t_0, s_1, t_1, \dots)$, that is, $\iota(s, t)_{2n} = s_n$ and $\iota(s, t)_{2n+1} = t_n$. This is

a map into $\{0, 1, \ldots, 9\}^{\mathbb{N}}$. It is injective, because s and t are recovered from $\iota(s, t)$ as its terms in the even positions $2n$ and the odd positions $2n + 1$ respectively. Its image lies in D_0. Indeed, $\iota(s, t)$ does not terminate with constant 9, because otherwise both s and t would have to terminate with constant 9. It is also not the zero sequence, because s is not.
This concludes the proof.